u (2)
Particle 0
pid: u (2)
status: 21
statusFlags: 14721
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 621.686, 621.686)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 701 simE=69.180542
subgraph recHits: 0 missing=701 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 54 dE=102.202339
raw GEN: GenParticle #94 key=3
raw SIM: n/a

production
Vertex 0
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 2 nOut: 1
raw GEN: GenVertex #3 key=-3
raw SIM: n/a

anti-u (-2)
Particle 1
pid: anti-u (-2)
status: 21
statusFlags: 14721
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, -2.023, 2.023)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 701 simE=69.180542
subgraph recHits: 0 missing=701 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 54 dE=102.202339
raw GEN: GenParticle #96 key=4
raw SIM: n/a

Z0 (23)
Particle 2
signal
pid: Z0 (23)
status: 62
statusFlags: 14721
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (13.716, -14.441, 568.482, 573.236)
nProdVtx: 1 nDecayVtx: 1
nParents: 2 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 701 simE=69.180542
subgraph recHits: 0 missing=701 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 54 dE=102.202339
raw GEN: GenParticle #134 key=36
raw SIM: n/a

decay
Vertex 1
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 1 nOut: 2
raw GEN: GenVertex #35 key=-35
raw SIM: n/a

tau- (15)
Particle 3
hardProcess
pid: tau- (15)
status: 23
statusFlags: 22913
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isFirstCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (16.585, 9.967, 43.113, 47.289)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 678 simE=2.180540
subgraph recHits: 0 missing=678 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 32 dE=68.202354
raw GEN: GenParticle #144 key=51
raw SIM: n/a

decay
Vertex 2
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 1 nOut: 2
raw GEN: GenVertex #36 key=-36
raw SIM: n/a

tau+ (-15)
Particle 4
hardProcess
tauVisibleHadronic
pid: tau+ (-15)
status: 2
statusFlags: 12675
flags: isPrompt, isDecayedLeptonHadron, isHardProcess, fromHardProcess, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-2.861, -24.341, 523.941, 524.517)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 23 simE=67.000000
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 22 dE=34.000000
raw GEN: GenParticle #148 key=55
raw SIM: n/a

Primary
Vertex 3
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.007, -0.064, 4.824, 0.244)
nIn: 1 nOut: 4
raw GEN: GenVertex #67 key=-67
raw SIM: SimVertex #297 key=1

tau- (15)
Particle 5
tauVisibleHadronic
pid: tau- (15)
status: 2
statusFlags: 8451
flags: isPrompt, isDecayedLeptonHadron, fromHardProcess, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (14.982, 9.711, 42.420, 46.059)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 568 simE=0.251752
subgraph recHits: 0 missing=568 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 19 dE=37.024998
raw GEN: GenParticle #146 key=53
raw SIM: n/a

Primary
Vertex 4
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.051, 0.033, 3.594, 0.203)
nIn: 1 nOut: 2
raw GEN: GenVertex #68 key=-68
raw SIM: SimVertex #298 key=2

gamma (22)
Particle 6
stableLegsFromInitialState
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (1.595, 0.189, 2.121, 2.660)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 110 simE=1.928787
subgraph recHits: 0 missing=110 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 13 dE=31.177345
raw GEN: GenParticle #147 key=54
raw SIM: SimTrack #501 key=1

PairConversion

Vertex 115
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (87.960, 10.439, 120.441, 5.092)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #455 key=159

pi- (-211)
Particle 7
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.195, 0.223, 97.546, 97.547)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #150 key=57
raw SIM: n/a

p (2212)
Particle 8
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: p (2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-3.758, 7.949, 24.736, 26.269)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 366 simE=0.130802
direct recHits: 0 missing=366 recoE=0.000000
subgraph simHits: 366 simE=0.130802
subgraph recHits: 0 missing=366 recoE=0.000000
direct tracker simHits: 33 dE=33.696850
subgraph tracker simHits: 33 dE=33.696850
checkpointId: 0
x4@checkpoint: (-37.845, 95.662, 293.500, 0.000)
p4@checkpoint: (-2.697, 8.338, 24.709, 26.234)
raw GEN: GenParticle #161 key=68
raw SIM: SimTrack #502 key=2

anti-p (-2212)
Particle 9
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: anti-p (-2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.377, 1.105, 3.284, 3.609)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 115 simE=0.215098
direct recHits: 0 missing=115 recoE=0.000000
subgraph simHits: 115 simE=0.215098
subgraph recHits: 0 missing=115 recoE=0.000000
direct tracker simHits: 41 dE=41.488979
subgraph tracker simHits: 41 dE=41.488979
checkpointId: 0
x4@checkpoint: (-73.281, 67.828, 293.500, 0.000)
p4@checkpoint: (-1.124, 0.305, 3.256, 3.583)
raw GEN: GenParticle #162 key=69
raw SIM: SimTrack #503 key=3

pi+ (211)
Particle 10
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.241, 1.814, 5.519, 5.816)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 124 simE=0.077296
direct recHits: 0 missing=124 recoE=0.000000
subgraph simHits: 124 simE=0.077296
subgraph recHits: 0 missing=124 recoE=0.000000
direct tracker simHits: 32 dE=32.772278
subgraph tracker simHits: 32 dE=32.772278
checkpointId: 0
x4@checkpoint: (16.295, 94.070, 293.500, 0.000)
p4@checkpoint: (0.816, 1.642, 5.484, 5.784)
raw GEN: GenParticle #163 key=70
raw SIM: SimTrack #504 key=4

pi- (-211)
Particle 11
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.580, 0.466, 1.836, 1.986)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 49 simE=0.030381
direct recHits: 0 missing=49 recoE=0.000000
subgraph simHits: 49 simE=0.030381
subgraph recHits: 0 missing=49 recoE=0.000000
direct tracker simHits: 44 dE=44.657604
subgraph tracker simHits: 44 dE=44.657604
checkpointId: 0
x4@checkpoint: (-98.091, -20.893, 293.500, 0.000)
p4@checkpoint: (-0.322, -0.615, 1.825, 1.957)
raw GEN: GenParticle #165 key=72
raw SIM: SimTrack #505 key=5

pi+ (211)
Particle 12
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.305, 0.991, 3.088, 3.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 94 simE=1.003051
direct recHits: 0 missing=94 recoE=0.000000
subgraph simHits: 94 simE=1.003051
subgraph recHits: 0 missing=94 recoE=0.000000
direct tracker simHits: 30 dE=30.036655
subgraph tracker simHits: 30 dE=30.036655
checkpointId: 0
x4@checkpoint: (-56.005, 109.847, 245.611, 0.000)
p4@checkpoint: (-0.051, 1.589, 3.046, 3.439)
raw GEN: GenParticle #167 key=74
raw SIM: SimTrack #506 key=6

pi+ (211)
Particle 13
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.077, 0.416, 0.841, 0.952)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.002404
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.002404
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 55 dE=55.681332
subgraph tracker simHits: 55 dE=55.681332
checkpointId: 0
x4@checkpoint: (62.716, -15.665, 293.500, 0.000)
p4@checkpoint: (-0.249, -0.296, 0.812, 0.910)
raw GEN: GenParticle #169 key=76
raw SIM: SimTrack #507 key=7

K- (-321)
Particle 14
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.161, -0.108, -3.043, 3.089)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 19 dE=31.000000
raw GEN: GenParticle #171 key=78
raw SIM: SimTrack #508 key=8

Decay
Vertex 114
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (0.554, -0.357, -6.891, 0.548)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #453 key=157

pdg(3122)
Particle 15
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pdg(3122)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.133, -0.110, -20.808, 20.839)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 3 dE=3.000000
raw GEN: GenParticle #174 key=81
raw SIM: SimTrack #509 key=9

Decay
Vertex 113
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (0.561, -0.465, -84.558, 3.138)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #452 key=156

K+ (321)
Particle 16
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.431, -0.076, -149.907, 149.908)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #177 key=84
raw SIM: n/a

pi- (-211)
Particle 17
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.227, -0.213, -32.332, 32.334)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=3.007682
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #187 key=94
raw SIM: SimTrack #510 key=10

HadronInelastic

Vertex 112
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-2.090, -2.218, -311.324, 10.698)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #445 key=149

pi+ (211)
Particle 18
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.451, 0.548, -44.914, 44.920)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=12.000000
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=12.000000
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (12.159, 12.755, -1114.950, 0.000)
p4@checkpoint: (0.490, 0.511, -44.898, 44.904)
raw GEN: GenParticle #188 key=95
raw SIM: SimTrack #511 key=11

pi- (-211)
Particle 19
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.306, -0.225, -187.578, 187.579)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #189 key=96
raw SIM: n/a

pi- (-211)
Particle 20
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.138, -0.788, -3.381, 3.477)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 81 simE=0.015736
direct recHits: 0 missing=81 recoE=0.000000
subgraph simHits: 81 simE=0.015736
subgraph recHits: 0 missing=81 recoE=0.000000
direct tracker simHits: 46 dE=47.003345
subgraph tracker simHits: 46 dE=47.003345
checkpointId: 0
x4@checkpoint: (21.824, -64.191, -293.500, 0.000)
p4@checkpoint: (0.580, -0.543, -3.347, 3.443)
raw GEN: GenParticle #193 key=100
raw SIM: SimTrack #512 key=12

K+ (321)
Particle 21
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.254, 0.038, -0.020, 0.557)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 67 dE=67.483978
subgraph tracker simHits: 26 dE=68.229774
raw GEN: GenParticle #196 key=103
raw SIM: SimTrack #513 key=13

Decay
Vertex 111
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.087, -9.806, 0.455, 2.232)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #444 key=148

K- (-321)
Particle 22
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.218, 0.001, -0.068, 0.544)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 6
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 32 simE=0.257210
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 21 dE=20.081810
subgraph tracker simHits: 10 dE=20.081812
raw GEN: GenParticle #197 key=104
raw SIM: SimTrack #514 key=14

HadronAtRest

Vertex 105
domain: SIM
reason: HadronAtRest
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.868, -13.458, -4.447, 2.477)
nIn: 1 nOut: 5
raw GEN: n/a
raw SIM: SimVertex #437 key=141

HadronInelastic

Vertex 106
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.825, -1.693, 4.453, 8.435)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #439 key=143

pi+ (211)
Particle 23
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.470, 0.166, 8.292, 8.308)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 25 dE=25.000000
subgraph tracker simHits: 25 dE=25.000000
checkpointId: 0
x4@checkpoint: (-14.998, 9.107, 293.500, 0.000)
p4@checkpoint: (-0.378, 0.347, 8.236, 8.253)
raw GEN: GenParticle #200 key=107
raw SIM: SimTrack #515 key=15

pi- (-211)
Particle 24
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.649, -0.090, 2.596, 2.681)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 50 simE=0.024047
direct recHits: 0 missing=50 recoE=0.000000
subgraph simHits: 50 simE=0.024047
subgraph recHits: 0 missing=50 recoE=0.000000
direct tracker simHits: 76 dE=74.021294
subgraph tracker simHits: 76 dE=74.021294
checkpointId: 0
x4@checkpoint: (59.735, 33.332, 293.500, 0.000)
p4@checkpoint: (0.268, 0.581, 2.547, 2.630)
raw GEN: GenParticle #201 key=108
raw SIM: SimTrack #516 key=16

pdg(3122)
Particle 25
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pdg(3122)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.509, 0.477, 43.792, 43.811)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=7.000000
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #202 key=109
raw SIM: SimTrack #517 key=17

Decay
Vertex 104
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-3.499, 3.279, 304.697, 10.251)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #434 key=138

anti-p (-2212)
Particle 26
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: anti-p (-2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.564, -0.178, 122.454, 122.459)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #204 key=111
raw SIM: n/a

pi+ (211)
Particle 27
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.093, 0.316, 275.443, 275.443)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #205 key=112
raw SIM: n/a

pi+ (211)
Particle 28
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-1.231, 0.306, 1.257, 1.791)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 106 simE=1.188681
subgraph recHits: 0 missing=106 recoE=0.000000
direct tracker simHits: 12 dE=12.000000
subgraph tracker simHits: 18 dE=39.098087
raw GEN: GenParticle #208 key=115
raw SIM: SimTrack #519 key=19

HadronInelastic

Vertex 84
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.710, 6.949, 24.218, 1.184)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #406 key=110

pi- (-211)
Particle 29
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.086, 0.929, 2.709, 3.066)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 84 simE=1.618957
direct recHits: 0 missing=84 recoE=0.000000
subgraph simHits: 84 simE=1.618957
subgraph recHits: 0 missing=84 recoE=0.000000
direct tracker simHits: 67 dE=67.862465
subgraph tracker simHits: 67 dE=67.862465
checkpointId: 0
x4@checkpoint: (-121.011, 23.649, 247.227, 0.000)
p4@checkpoint: (-1.328, -0.412, 2.657, 3.002)
raw GEN: GenParticle #209 key=116
raw SIM: SimTrack #520 key=20

pi+ (211)
Particle 30
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.665, -0.045, 0.646, 0.939)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.001920
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 41 dE=41.318695
subgraph tracker simHits: 19 dE=51.810844
raw GEN: GenParticle #210 key=117
raw SIM: SimTrack #521 key=21

HadronInelastic

Vertex 83
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.839, 112.887, 196.058, 9.527)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #405 key=109

pi- (-211)
Particle 31
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.078, 0.151, 0.221, 0.312)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002660
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002660
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 40 dE=40.100307
subgraph tracker simHits: 40 dE=40.100307
checkpointId: 0
x4@checkpoint: (-21.171, -2.196, 293.500, 0.000)
p4@checkpoint: (0.116, -0.059, 0.177, 0.260)
raw GEN: GenParticle #211 key=118
raw SIM: SimTrack #522 key=22

n (2112)
Particle 32
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: n (2112)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-1.125, 0.286, 1.857, 2.383)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 45 simE=0.755054
direct recHits: 0 missing=45 recoE=0.000000
subgraph simHits: 46 simE=0.755072
subgraph recHits: 0 missing=46 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.461, 30.529, 198.783, 0.000)
p4@checkpoint: (-1.277, 0.358, 1.741, 2.381)
raw GEN: GenParticle #212 key=119
raw SIM: SimTrack #523 key=23

HadronInelastic

Vertex 82
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-118.743, -11.495, 212.805, 18.867)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #404 key=108

pi- (-211)
Particle 33
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-1.015, 0.097, 1.350, 1.698)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 45 simE=0.621877
direct recHits: 0 missing=45 recoE=0.000000
subgraph simHits: 49 simE=0.626534
subgraph recHits: 0 missing=49 recoE=0.000000
direct tracker simHits: 33 dE=34.281620
subgraph tracker simHits: 18 dE=34.281616
checkpointId: 0
x4@checkpoint: (-97.343, -75.679, 182.923, 0.000)
p4@checkpoint: (-0.153, -0.992, 1.294, 1.644)
raw GEN: GenParticle #213 key=120
raw SIM: SimTrack #524 key=24

HadronInelastic

Vertex 81
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (40.313, 109.310, 99.823, 55.356)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #403 key=107

pdg(-3122)
Particle 34
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pdg(-3122)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.179, 0.387, 0.042, 1.195)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 85 simE=0.266532
subgraph recHits: 0 missing=85 recoE=0.000000
direct tracker simHits: 63 dE=59.590282
subgraph tracker simHits: 37 dE=110.335487
raw GEN: GenParticle #214 key=121
raw SIM: SimTrack #525 key=25

HadronAtRest

Vertex 53
domain: SIM
reason: HadronAtRest
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.230, 14.051, 5.039, 2.242)
nIn: 1 nOut: 5
raw GEN: n/a
raw SIM: SimVertex #371 key=75

pi+ (211)
Particle 35
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.154, -0.176, -0.122, 0.299)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.063462
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 11 dE=11.000000
subgraph tracker simHits: 4 dE=11.907351
raw GEN: GenParticle #216 key=123
raw SIM: SimTrack #526 key=26

HadronInelastic

Vertex 51
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.484, -6.211, -8.672, 1.203)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #369 key=73

pi- (-211)
Particle 36
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.270, 0.024, -1.049, 1.093)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.001591
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.001591
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 44 dE=44.521076
subgraph tracker simHits: 44 dE=44.521076
checkpointId: 0
x4@checkpoint: (-2.631, -48.620, -293.500, 0.000)
p4@checkpoint: (0.253, 0.001, -1.013, 1.054)
raw GEN: GenParticle #217 key=124
raw SIM: SimTrack #527 key=27

K0_S (310)
Particle 37
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_S (310)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.401, -0.336, -200.717, 200.718)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #218 key=125
raw SIM: n/a

n (2112)
Particle 38
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: n (2112)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.335, -0.294, -609.027, 609.028)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #219 key=126
raw SIM: n/a

pi+ (211)
Particle 39
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.263, 0.071, -167.177, 167.178)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #221 key=128
raw SIM: n/a

pi+ (211)
Particle 40
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.496, -0.562, -237.910, 237.912)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #223 key=130
raw SIM: n/a

pi- (-211)
Particle 41
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.247, 0.061, -24.756, 24.758)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (11.688, 4.419, -1218.281, 0.000)
p4@checkpoint: (0.238, 0.086, -24.697, 24.699)
raw GEN: GenParticle #224 key=131
raw SIM: SimTrack #528 key=28

K+ (321)
Particle 42
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.274, 0.205, -17.688, 17.698)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=6.000000
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=6.000000
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.478, 4.043, -430.296, 0.000)
p4@checkpoint: (0.326, 0.127, -17.674, 17.685)
raw GEN: GenParticle #225 key=132
raw SIM: SimTrack #529 key=29

K0_L (130)
Particle 43
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K0_L (130)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.301, 0.271, -7.099, 7.128)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-12.574, 11.350, -293.500, 0.000)
p4@checkpoint: (-0.301, 0.271, -7.099, 7.128)
raw GEN: GenParticle #227 key=134
raw SIM: SimTrack #530 key=30

pi+ (211)
Particle 44
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.052, 0.394, 4.337, 4.357)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 17
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 148 simE=1.274176
subgraph recHits: 0 missing=148 recoE=0.000000
direct tracker simHits: 149 dE=139.027802
subgraph tracker simHits: 78 dE=290.611115
raw GEN: GenParticle #228 key=135
raw SIM: SimTrack #531 key=31

HadronInelastic

Vertex 43
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-0.244, 5.846, 67.574, 2.347)
nIn: 1 nOut: 12
raw GEN: n/a
raw SIM: SimVertex #360 key=64

NuclearCapture
Vertex 44
domain: SIM
reason: NuclearCapture
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-115.318, 15.631, 199.325, 83.784)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #361 key=65

HadronInelastic

Vertex 45
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-32.355, -62.235, 106.763, 25.303)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #363 key=67

Bremsstrahlung
Vertex 46
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.412, 5.836, 40.892, 484.491)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #364 key=68

Annihilation

Vertex 47
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (0.978, 5.880, 110.546, 487.524)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #365 key=69

pi- (-211)
Particle 45
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.435, -0.019, 2.976, 3.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 38 simE=0.005315
direct recHits: 0 missing=38 recoE=0.000000
subgraph simHits: 38 simE=0.005315
subgraph recHits: 0 missing=38 recoE=0.000000
direct tracker simHits: 21 dE=21.145123
subgraph tracker simHits: 21 dE=21.145123
checkpointId: 0
x4@checkpoint: (34.712, 21.022, 293.500, 0.000)
p4@checkpoint: (0.179, 0.398, 2.925, 2.961)
raw GEN: GenParticle #229 key=136
raw SIM: SimTrack #532 key=32

K+ (321)
Particle 46
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.017, 0.119, 15.967, 15.975)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #230 key=137
raw SIM: n/a

pi+ (211)
Particle 47
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.182, 0.690, 1570.937, 1570.937)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #232 key=139
raw SIM: n/a

anti-nu_tau (-16)
Particle 48
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 13884
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isHardProcessTauDecayProduct, isDirectHardProcessTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-2.448, -18.267, 391.230, 391.663)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #234 key=141
raw SIM: n/a

pi+ (211)
Particle 49
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 13884
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isHardProcessTauDecayProduct, isDirectHardProcessTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.295, -2.183, 46.530, 46.583)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=27.000000
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=27.000000
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 10 dE=10.000000
subgraph tracker simHits: 10 dE=10.000000
checkpointId: 0
x4@checkpoint: (-2.319, -13.508, 293.500, 0.000)
p4@checkpoint: (-0.447, -2.152, 46.515, 46.567)
raw GEN: GenParticle #235 key=142
raw SIM: SimTrack #533 key=34

pi+ (211)
Particle 50
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 13884
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isHardProcessTauDecayProduct, isDirectHardProcessTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.145, -1.445, 29.047, 29.083)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=13.000000
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=13.000000
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 14 dE=14.000000
subgraph tracker simHits: 14 dE=14.000000
checkpointId: 0
x4@checkpoint: (0.609, -14.524, 293.500, 0.000)
p4@checkpoint: (-0.016, -1.465, 29.029, 29.066)
raw GEN: GenParticle #236 key=143
raw SIM: SimTrack #534 key=35

pi- (-211)
Particle 51
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 13884
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isHardProcessTauDecayProduct, isDirectHardProcessTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.263, -2.446, 57.134, 57.187)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=27.000000
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=27.000000
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 10 dE=10.000000
subgraph tracker simHits: 10 dE=10.000000
checkpointId: 0
x4@checkpoint: (-1.017, -12.468, 293.500, 0.000)
p4@checkpoint: (-0.140, -2.459, 57.104, 57.158)
raw GEN: GenParticle #237 key=144
raw SIM: SimTrack #535 key=36

nu_tau (16)
Particle 52
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 13884
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isHardProcessTauDecayProduct, isDirectHardProcessTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (3.106, 1.307, 8.923, 9.538)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #238 key=145
raw SIM: n/a

pi- (-211)
Particle 53
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 13884
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isHardProcessTauDecayProduct, isDirectHardProcessTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (11.876, 8.404, 33.497, 36.520)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 566 simE=0.251079
direct recHits: 0 missing=566 recoE=0.000000
subgraph simHits: 568 simE=0.251752
subgraph recHits: 0 missing=568 recoE=0.000000
direct tracker simHits: 38 dE=37.024998
subgraph tracker simHits: 19 dE=37.024998
checkpointId: 0
x4@checkpoint: (97.169, 75.902, 287.476, 0.000)
p4@checkpoint: (11.026, 9.465, 33.447, 36.467)
raw GEN: GenParticle #239 key=146
raw SIM: SimTrack #536 key=38

Bremsstrahlung
Vertex 41
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (109.160, 16.079, 246.948, 15.928)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #356 key=60

Bremsstrahlung
Vertex 42
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.243, 9.965, 43.524, 1.657)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #357 key=61

pi+ (211)
Particle 54
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.135, -0.431, 919.635, 919.635)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #240 key=147
raw SIM: n/a

pi- (-211)
Particle 55
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.058, -0.183, 200.688, 200.689)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #241 key=148
raw SIM: n/a

pi+ (211)
Particle 56
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.030, -0.185, -1.966, 1.980)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000130
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000130
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 21 dE=21.000000
subgraph tracker simHits: 21 dE=21.000000
checkpointId: 0
x4@checkpoint: (-17.623, -14.098, -293.500, 0.000)
p4@checkpoint: (-0.156, 0.028, -1.950, 1.961)
raw GEN: GenParticle #243 key=150
raw SIM: SimTrack #537 key=39

pi- (-211)
Particle 57
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.186, 0.114, -0.539, 0.598)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.001199
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 122 dE=118.795906
subgraph tracker simHits: 20 dE=118.795906
raw GEN: GenParticle #244 key=151
raw SIM: SimTrack #538 key=40

HadronInelastic

Vertex 40
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-26.455, 15.159, -192.332, 7.456)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #355 key=59

pdg(-3122)
Particle 58
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pdg(-3122)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.209, 0.335, -20.811, 20.844)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=4.000000
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #246 key=153
raw SIM: SimTrack #539 key=41

Decay
Vertex 39
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.156, 3.457, -211.156, 7.368)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #351 key=55

gamma (22)
Particle 59
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.232, -0.085, -52.279, 52.280)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #248 key=155
raw SIM: n/a

gamma (22)
Particle 60
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.259, -0.164, -100.563, 100.564)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #249 key=156
raw SIM: n/a

gamma (22)
Particle 61
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.357, -0.185, -368.524, 368.524)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #250 key=157
raw SIM: n/a

gamma (22)
Particle 62
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.229, -0.201, -377.238, 377.238)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #251 key=158
raw SIM: n/a

pi+ (211)
Particle 63
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.245, 0.009, -473.887, 473.887)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #252 key=159
raw SIM: n/a

pi- (-211)
Particle 64
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.037, 0.156, -1261.813, 1261.813)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #253 key=160
raw SIM: n/a

pi+ (211)
Particle 65
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.071, -0.421, -20.001, 20.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=3.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=3.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (0.549, -8.265, -381.592, 0.000)
p4@checkpoint: (-0.014, -0.431, -19.989, 19.994)
raw GEN: GenParticle #258 key=165
raw SIM: SimTrack #541 key=43

pi- (-211)
Particle 66
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.158, -0.014, -9.551, 9.553)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=3.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=3.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-15.608, -9.018, -1114.950, 0.000)
p4@checkpoint: (-0.129, -0.078, -9.536, 9.538)
raw GEN: GenParticle #259 key=166
raw SIM: SimTrack #542 key=44

gamma (22)
Particle 67
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.320, 0.493, -1.755, 1.851)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 34 simE=0.008340
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 18 dE=26.259140
raw GEN: GenParticle #261 key=168
raw SIM: SimTrack #543 key=45

PairConversion

Vertex 36
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-24.982, 38.497, -133.535, 5.017)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #343 key=47

gamma (22)
Particle 68
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.039, 0.100, -0.535, 0.546)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 15 simE=0.002086
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-21.871, 55.626, -293.500, 0.000)
p4@checkpoint: (-0.039, 0.100, -0.535, 0.546)
raw GEN: GenParticle #262 key=169
raw SIM: SimTrack #544 key=46

PairConversion

Vertex 35
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.115, 58.790, -310.391, 10.876)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #342 key=46

gamma (22)
Particle 69
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.058, -0.139, 0.111, 0.187)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 11 simE=0.175704
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.682, -113.707, 94.202, 0.000)
p4@checkpoint: (0.058, -0.139, 0.111, 0.187)
raw GEN: GenParticle #263 key=170
raw SIM: SimTrack #545 key=47

PairConversion

Vertex 34
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (50.101, -119.476, 98.806, 5.564)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #341 key=45

gamma (22)
Particle 70
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.124, -0.062, 0.009, 0.139)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.054537
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=2.092179
raw GEN: GenParticle #264 key=171
raw SIM: SimTrack #546 key=48

PairConversion

Vertex 25
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.466, -51.760, 10.916, 4.065)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #330 key=34

pi+ (211)
Particle 71
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.100, -0.220, 1.046, 1.083)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.004699
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 30 simE=0.004699
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 39 dE=39.671642
subgraph tracker simHits: 39 dE=39.671642
checkpointId: 0
x4@checkpoint: (-39.317, 18.553, 293.500, 0.000)
p4@checkpoint: (0.097, 0.208, 1.018, 1.052)
raw GEN: GenParticle #265 key=172
raw SIM: SimTrack #547 key=49

pi- (-211)
Particle 72
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.164, 0.071, 0.657, 0.695)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 21 dE=77.956657
raw GEN: GenParticle #266 key=173
raw SIM: SimTrack #548 key=50

Decay
Vertex 24
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (0.612, 0.279, 5.917, 0.285)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #328 key=32

gamma (22)
Particle 73
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.074, -0.044, 0.061, 0.106)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 10 simE=0.020480
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=51.225040
raw GEN: GenParticle #268 key=175
raw SIM: SimTrack #549 key=51

PairConversion

Vertex 18
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.644, -12.272, 20.442, 1.179)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #322 key=26

gamma (22)
Particle 74
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.222, 0.088, 0.081, 0.252)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 14 simE=0.248445
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-114.695, 45.255, 45.356, 0.000)
p4@checkpoint: (-0.222, 0.088, 0.081, 0.252)
raw GEN: GenParticle #269 key=176
raw SIM: SimTrack #550 key=52

PairConversion

Vertex 17
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-121.782, 48.051, 47.945, 4.810)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #321 key=25

gamma (22)
Particle 75
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.084, 0.173, -202.691, 202.691)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #270 key=177
raw SIM: n/a

gamma (22)
Particle 76
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.226, 0.126, -292.335, 292.335)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #271 key=178
raw SIM: n/a

gamma (22)
Particle 77
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.298, -0.037, -464.088, 464.089)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #272 key=179
raw SIM: n/a

gamma (22)
Particle 78
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.032, -0.062, -115.605, 115.605)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #273 key=180
raw SIM: n/a

gamma (22)
Particle 79
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.069, 0.369, -22.726, 22.729)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=6.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=6.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (3.393, 18.165, -1114.950, 0.000)
p4@checkpoint: (0.069, 0.369, -22.726, 22.729)
raw GEN: GenParticle #274 key=181
raw SIM: SimTrack #551 key=53

gamma (22)
Particle 80
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.016, 0.029, -5.227, 5.227)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #275 key=182
raw SIM: n/a

gamma (22)
Particle 81
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.281, -0.163, 4.060, 4.073)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.000975
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (20.096, -11.650, 293.500, 0.000)
p4@checkpoint: (0.281, -0.163, 4.060, 4.073)
raw GEN: GenParticle #276 key=183
raw SIM: SimTrack #552 key=54

PairConversion

Vertex 16
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (23.052, -13.364, 336.170, 11.332)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #320 key=24

gamma (22)
Particle 82
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.008, -0.040, 6.609, 6.609)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #280 key=187
raw SIM: n/a

gamma (22)
Particle 83
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.091, -0.117, 98.862, 98.862)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #281 key=188
raw SIM: n/a

gamma (22)
Particle 84
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.121, -0.049, -0.383, 0.404)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 10 simE=0.002008
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-94.025, -37.702, -293.500, 0.000)
p4@checkpoint: (-0.121, -0.049, -0.383, 0.404)
raw GEN: GenParticle #282 key=189
raw SIM: SimTrack #556 key=58

PairConversion

Vertex 15
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-103.155, -41.362, -322.334, 11.680)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #319 key=23

gamma (22)
Particle 85
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.011, -0.040, -0.410, 0.412)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.000681
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.961, -29.252, -293.500, 0.000)
p4@checkpoint: (0.011, -0.040, -0.410, 0.412)
raw GEN: GenParticle #283 key=190
raw SIM: SimTrack #557 key=59

PairConversion

Vertex 14
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.322, -30.579, -306.966, 10.606)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #318 key=22

gamma (22)
Particle 86
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.058, 0.052, -121.504, 121.504)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #285 key=192
raw SIM: n/a

gamma (22)
Particle 87
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.144, 0.004, -36.498, 36.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #286 key=193
raw SIM: n/a

gamma (22)
Particle 88
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.000, 0.059, -9.292, 9.292)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #287 key=194
raw SIM: n/a

gamma (22)
Particle 89
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.053, 0.104, -43.197, 43.197)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #288 key=195
raw SIM: n/a

gamma (22)
Particle 90
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.067, 0.111, -16.906, 16.906)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #289 key=196
raw SIM: n/a

gamma (22)
Particle 91
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.078, -0.056, -24.738, 24.738)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #290 key=197
raw SIM: n/a

gamma (22)
Particle 92
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.004, 0.057, -11.494, 11.494)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #291 key=198
raw SIM: n/a

gamma (22)
Particle 93
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.070, 0.061, -13.962, 13.962)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #293 key=200
raw SIM: n/a

gamma (22)
Particle 94
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.072, -0.222, 0.424, 0.484)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 20 simE=0.017018
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 7 dE=16.316195
raw GEN: GenParticle #294 key=201
raw SIM: SimTrack #560 key=62

PairConversion

Vertex 6
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-30.601, -94.874, 184.478, 7.091)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #304 key=8

gamma (22)
Particle 95
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (0.015, -0.207, 0.632, 0.665)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 25 simE=0.005141
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.065, -95.036, 293.500, 0.000)
p4@checkpoint: (0.015, -0.207, 0.632, 0.665)
raw GEN: GenParticle #295 key=202
raw SIM: SimTrack #561 key=63

PairConversion

Vertex 5
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.740, -104.117, 321.213, 11.355)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #303 key=7

e- (11)
Particle 96
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, -0.064, 0.196, 0.206)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.001789
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.001789
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.743, -104.147, 321.312, 0.000)
p4@checkpoint: (0.010, -0.056, 0.196, 0.204)
raw GEN: n/a
raw SIM: SimTrack #562 key=64

e+ (-11)
Particle 97
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.012, -0.143, 0.436, 0.459)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 20 simE=0.003352
direct recHits: 0 missing=20 recoE=0.000000
subgraph simHits: 20 simE=0.003352
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.741, -104.150, 321.314, 0.000)
p4@checkpoint: (0.010, -0.142, 0.435, 0.457)
raw GEN: n/a
raw SIM: SimTrack #563 key=65

e- (11)
Particle 98
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.027, -0.084, 0.163, 0.185)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000243
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000243
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 9 dE=8.919325
subgraph tracker simHits: 9 dE=8.919325
checkpointId: 0
x4@checkpoint: (-26.918, -105.075, 293.500, 0.000)
p4@checkpoint: (0.068, -0.043, 0.155, 0.175)
raw GEN: n/a
raw SIM: SimTrack #564 key=1229

e+ (-11)
Particle 99
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.044, -0.137, 0.262, 0.299)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 7
nCheckpoints: 1
direct simHits: 7 simE=0.002376
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 17 simE=0.016775
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 6 dE=6.342458
subgraph tracker simHits: 3 dE=7.396867
checkpointId: 0
x4@checkpoint: (-47.776, -103.622, 293.500, 0.000)
p4@checkpoint: (-0.054, 0.014, 0.119, 0.132)
raw GEN: n/a
raw SIM: SimTrack #565 key=1230

Bremsstrahlung
Vertex 7
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-38.579, -103.359, 207.337, 7.954)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #305 key=9

Bremsstrahlung
Vertex 8
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.120, -103.849, 220.158, 8.435)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #306 key=10

Bremsstrahlung
Vertex 9
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.265, -103.820, 220.456, 8.446)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #307 key=11

Annihilation

Vertex 10
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.749, -103.997, 223.276, 8.586)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #311 key=15

Bremsstrahlung
Vertex 11
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-47.449, -105.288, 226.050, 8.661)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #312 key=16

Bremsstrahlung
Vertex 12
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-47.346, -105.506, 226.266, 8.672)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #313 key=17

Bremsstrahlung
Vertex 13
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-47.175, -105.383, 225.853, 8.688)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #315 key=19

gamma (22)
Particle 100
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.054409
subgraph tracker simHits: 1 dE=1.054409
checkpointId: 0
x4@checkpoint: (-24.011, 32.596, 293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #566 key=1233

gamma (22)
Particle 101
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, 0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000104
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000104
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-64.574, -70.139, 293.500, 0.000)
p4@checkpoint: (-0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #567 key=1235

gamma (22)
Particle 102
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, 0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001247
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001247
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-70.740, -100.989, 232.507, 0.000)
p4@checkpoint: (-0.001, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #568 key=1238

gamma (22)
Particle 103
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000322
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000322
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (41.520, -116.099, -123.579, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #572 key=1491

gamma (22)
Particle 104
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.008, 0.009, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.011742
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.011742
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.438, -113.387, 234.429, 0.000)
p4@checkpoint: (-0.001, -0.008, 0.009, 0.012)
raw GEN: n/a
raw SIM: SimTrack #574 key=1498

gamma (22)
Particle 105
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000062
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000062
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.183, -113.496, 221.640, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #575 key=1500

gamma (22)
Particle 106
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000922
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000922
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-109.097, -57.452, 201.770, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #577 key=1506

e- (11)
Particle 107
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, -0.031, -0.313, 0.315)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.000681
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.000681
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (8.337, -30.623, -307.412, 0.000)
p4@checkpoint: (0.013, -0.031, -0.312, 0.314)
raw GEN: n/a
raw SIM: SimTrack #580 key=1795

e+ (-11)
Particle 108
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.118, -0.046, -0.373, 0.394)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.002008
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.002008
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-103.159, -41.364, -322.346, 0.000)
p4@checkpoint: (-0.038, -0.015, -0.124, 0.130)
raw GEN: n/a
raw SIM: SimTrack #583 key=2451

e+ (-11)
Particle 109
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.258, -0.150, 3.722, 3.734)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.000975
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.000975
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (23.071, -13.375, 336.444, 0.000)
p4@checkpoint: (0.260, -0.149, 3.703, 3.715)
raw GEN: n/a
raw SIM: SimTrack #585 key=3118

e- (11)
Particle 110
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.077, 0.031, 0.028, 0.088)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.085261
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.085261
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.887, 48.086, 47.982, 0.000)
p4@checkpoint: (-0.079, 0.022, 0.025, 0.086)
raw GEN: n/a
raw SIM: SimTrack #586 key=6319

e+ (-11)
Particle 111
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.145, 0.057, 0.052, 0.164)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.163184
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.163184
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.822, 48.067, 47.960, 0.000)
p4@checkpoint: (-0.143, 0.055, 0.056, 0.163)
raw GEN: n/a
raw SIM: SimTrack #587 key=6320

e- (11)
Particle 112
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.048, -0.029, 0.039, 0.068)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.011451
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 11 dE=11.044383
subgraph tracker simHits: 4 dE=11.044382
raw GEN: n/a
raw SIM: SimTrack #588 key=6474

Bremsstrahlung
Vertex 21
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.700, -12.305, 20.489, 1.182)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #325 key=29

Bremsstrahlung
Vertex 22
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.214, -13.113, 53.126, 3.541)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #326 key=30

Bremsstrahlung
Vertex 23
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.427, -12.452, 47.260, 3.017)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #327 key=31

e+ (-11)
Particle 113
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.027, -0.014, 0.023, 0.038)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.009029
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 39 dE=40.180660
subgraph tracker simHits: 12 dE=40.180656
raw GEN: n/a
raw SIM: SimTrack #589 key=6475

Bremsstrahlung
Vertex 19
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-22.970, -7.578, 38.029, 2.203)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #323 key=27

Annihilation

Vertex 20
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-22.709, -9.351, 61.401, 3.696)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #324 key=28

gamma (22)
Particle 114
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.002, 0.005, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.008518
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.008518
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.180, 27.562, 132.235, 0.000)
p4@checkpoint: (0.007, 0.002, 0.005, 0.009)
raw GEN: n/a
raw SIM: SimTrack #590 key=6481

gamma (22)
Particle 115
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000511
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000511
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-32.718, -118.880, -64.341, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #591 key=6487

gamma (22)
Particle 116
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, -0.004, 0.006, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.009932
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.009932
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-105.382, -64.012, 88.539, 0.000)
p4@checkpoint: (-0.007, -0.004, 0.006, 0.010)
raw GEN: n/a
raw SIM: SimTrack #592 key=6499

gamma (22)
Particle 117
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, 0.011, 0.006, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000644
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000644
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.354, 81.693, 103.325, 0.000)
p4@checkpoint: (-0.000, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #593 key=6506

gamma (22)
Particle 118
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000875
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000875
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-67.147, -103.413, 8.946, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #594 key=6518

mu- (13)
Particle 119
caloBoundary
pid: mu- (13)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.172, 0.072, 0.609, 0.646)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 78 dE=77.956657
subgraph tracker simHits: 78 dE=77.956657
checkpointId: 0
x4@checkpoint: (6.592, 4.706, 340.381, 0.000)
p4@checkpoint: (0.128, 0.112, 0.537, 0.573)
raw GEN: n/a
raw SIM: SimTrack #595 key=6526

e- (11)
Particle 120
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.084, -0.042, 0.006, 0.094)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 10 simE=0.051074
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #597 key=7650

Bremsstrahlung
Vertex 31
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.605, -51.829, 10.930, 4.070)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #338 key=42

Bremsstrahlung
Vertex 32
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.055, -50.350, 7.365, 4.576)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #339 key=43

Bremsstrahlung
Vertex 33
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.086, -50.373, 7.348, 4.578)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #340 key=44

e+ (-11)
Particle 121
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.039, -0.021, 0.001, 0.045)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.003463
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=2.092179
raw GEN: n/a
raw SIM: SimTrack #598 key=7651

Bremsstrahlung
Vertex 26
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (100.645, -57.297, 10.396, 4.421)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #331 key=35

Bremsstrahlung
Vertex 27
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (102.613, -53.238, 9.486, 4.619)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #332 key=36

Bremsstrahlung
Vertex 28
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.477, -55.156, 8.303, 4.825)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #334 key=38

Bremsstrahlung
Vertex 29
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.712, -55.031, 8.415, 4.837)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #336 key=40

Annihilation

Vertex 30
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.596, -55.128, 8.332, 4.845)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #337 key=41

gamma (22)
Particle 122
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000359
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000359
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.740, 95.705, -47.383, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #599 key=7655

gamma (22)
Particle 123
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.000, -0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001520
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001520
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (111.063, -53.554, 8.159, 0.000)
p4@checkpoint: (0.002, -0.000, -0.000, 0.002)
raw GEN: n/a
raw SIM: SimTrack #600 key=7662

gamma (22)
Particle 124
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=2.092179
subgraph tracker simHits: 2 dE=2.092179
checkpointId: 0
x4@checkpoint: (105.028, -64.590, 273.314, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #602 key=7671

gamma (22)
Particle 125
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001227
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001227
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.146, -51.245, -0.663, 0.000)
p4@checkpoint: (0.001, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #604 key=7675

gamma (22)
Particle 126
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000357
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000357
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (116.525, -40.308, 56.535, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #605 key=7677

gamma (22)
Particle 127
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.048, -0.024, 0.006, 0.054)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.030484
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.030484
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.284, -55.139, 11.788, 0.000)
p4@checkpoint: (0.048, -0.024, 0.006, 0.054)
raw GEN: n/a
raw SIM: SimTrack #606 key=7685

gamma (22)
Particle 128
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000378
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000378
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (111.637, -52.346, 7.705, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #607 key=7692

gamma (22)
Particle 129
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.016, -0.009, -0.009, 0.020)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.020212
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.020212
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.710, -54.279, 3.432, 0.000)
p4@checkpoint: (0.016, -0.009, -0.009, 0.020)
raw GEN: n/a
raw SIM: SimTrack #608 key=7693

e- (11)
Particle 130
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.031, -0.075, 0.060, 0.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.095878
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.095878
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.209, -119.708, 99.022, 0.000)
p4@checkpoint: (0.033, -0.064, 0.061, 0.094)
raw GEN: n/a
raw SIM: SimTrack #609 key=7741

e+ (-11)
Particle 131
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.027, -0.064, 0.051, 0.086)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.079827
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.079827
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.122, -119.529, 98.850, 0.000)
p4@checkpoint: (0.025, -0.061, 0.050, 0.083)
raw GEN: n/a
raw SIM: SimTrack #610 key=7742

e- (11)
Particle 132
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.028, 0.070, -0.376, 0.383)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.001624
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 11 simE=0.001624
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-23.318, 59.224, -312.755, 0.000)
p4@checkpoint: (-0.037, 0.064, -0.372, 0.379)
raw GEN: n/a
raw SIM: SimTrack #611 key=7881

e+ (-11)
Particle 133
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.012, 0.030, -0.159, 0.163)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.000461
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.000461
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-23.153, 58.907, -310.987, 0.000)
p4@checkpoint: (-0.007, 0.034, -0.158, 0.161)
raw GEN: n/a
raw SIM: SimTrack #612 key=7882

e- (11)
Particle 134
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.038, 0.059, -0.210, 0.221)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000176
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000176
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 12 dE=11.813403
subgraph tracker simHits: 12 dE=11.813403
checkpointId: 0
x4@checkpoint: (-34.155, 33.233, -293.500, 0.000)
p4@checkpoint: (0.025, -0.034, -0.174, 0.179)
raw GEN: n/a
raw SIM: SimTrack #613 key=8788

e+ (-11)
Particle 135
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.282, 0.434, -1.546, 1.630)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 27 simE=0.007105
direct recHits: 0 missing=27 recoE=0.000000
subgraph simHits: 32 simE=0.008163
subgraph recHits: 0 missing=32 recoE=0.000000
direct tracker simHits: 14 dE=14.445736
subgraph tracker simHits: 11 dE=14.445737
checkpointId: 0
x4@checkpoint: (-23.901, 88.895, -293.500, 0.000)
p4@checkpoint: (0.252, 0.342, -1.278, 1.347)
raw GEN: n/a
raw SIM: SimTrack #614 key=8789

Bremsstrahlung
Vertex 37
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-25.109, 38.693, -134.232, 5.041)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #344 key=48

Bremsstrahlung
Vertex 38
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-28.194, 81.541, -267.971, 9.746)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #346 key=50

gamma (22)
Particle 136
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.016, 0.024, -0.086, 0.091)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000330
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000330
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-54.108, 83.748, -293.500, 0.000)
p4@checkpoint: (-0.016, 0.024, -0.086, 0.091)
raw GEN: n/a
raw SIM: SimTrack #615 key=8790

gamma (22)
Particle 137
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.024, 0.053, -0.174, 0.184)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000729
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000729
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.637, 89.332, -293.500, 0.000)
p4@checkpoint: (0.024, 0.053, -0.174, 0.184)
raw GEN: n/a
raw SIM: SimTrack #617 key=8792

anti-p (-2212)
Particle 138
caloBoundary
pid: anti-p (-2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.212, 0.335, -16.540, 16.572)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=4.000000
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=4.000000
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (4.214, 7.163, -387.687, 0.000)
p4@checkpoint: (0.174, 0.358, -16.540, 16.571)
raw GEN: n/a
raw SIM: SimTrack #623 key=11680

n (2112)
Particle 139
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.080, 0.125, -0.201, 0.972)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000761
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000761
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-66.906, 78.176, -293.500, 0.000)
p4@checkpoint: (-0.080, 0.125, -0.201, 0.972)
raw GEN: n/a
raw SIM: SimTrack #628 key=11748

n (2112)
Particle 140
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.178, -0.077, -0.108, 0.966)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000238
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000238
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-114.097, -46.741, -264.133, 0.000)
p4@checkpoint: (-0.148, -0.109, -0.125, 0.965)
raw GEN: n/a
raw SIM: SimTrack #629 key=11749

gamma (22)
Particle 141
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, 0.002, -0.003, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000200
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000200
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.515, 30.318, -220.464, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #631 key=11751

gamma (22)
Particle 142
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000313
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000313
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.779, -29.254, 215.036, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #633 key=26804

gamma (22)
Particle 143
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000360
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000360
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.195, 81.872, -55.209, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #634 key=62718

pi- (-211)
Particle 144
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.527, 0.115, 0.787, 0.965)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.075251
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 30 simE=0.075251
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 36 dE=36.779507
subgraph tracker simHits: 36 dE=36.779507
checkpointId: 0
x4@checkpoint: (-16.331, -86.400, 293.500, 0.000)
p4@checkpoint: (0.500, -0.063, 0.778, 0.937)
raw GEN: n/a
raw SIM: SimTrack #637 key=65565

pi+ (211)
Particle 145
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.303, 0.202, 0.752, 0.847)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.002999
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.002999
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 28 dE=29.001429
subgraph tracker simHits: 28 dE=29.001429
checkpointId: 0
x4@checkpoint: (23.664, -51.802, 293.500, 0.000)
p4@checkpoint: (-0.342, -0.078, 0.723, 0.816)
raw GEN: n/a
raw SIM: SimTrack #638 key=65566

pi0 (111)
Particle 146
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.675, -0.208, 0.219, 0.752)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 31 simE=0.797198
subgraph recHits: 0 missing=31 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #639 key=65567

Decay
Vertex 50
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-0.244, 5.846, 67.574, 2.347)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #368 key=72

pi- (-211)
Particle 147
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.411, 0.018, 0.056, 0.438)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 22 simE=0.378792
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 54 dE=53.930809
subgraph tracker simHits: 5 dE=53.930820
raw GEN: n/a
raw SIM: SimTrack #640 key=65568

HadronInelastic

Vertex 48
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.487, -51.504, 88.014, 7.249)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #366 key=70

p (2212)
Particle 148
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.151, 0.063, 0.679, 1.170)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 18 dE=18.000000
subgraph tracker simHits: 18 dE=18.000000
checkpointId: 0
x4@checkpoint: (2.494, -19.710, 293.500, 0.000)
p4@checkpoint: (-0.120, 0.044, 0.589, 1.115)
raw GEN: n/a
raw SIM: SimTrack #641 key=65569

p (2212)
Particle 149
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.120, 0.157, 0.646, 1.156)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 13 dE=12.725030
subgraph tracker simHits: 13 dE=12.725030
checkpointId: 0
x4@checkpoint: (11.510, -11.663, 293.500, 0.000)
p4@checkpoint: (-0.088, 0.068, 0.464, 1.053)
raw GEN: n/a
raw SIM: SimTrack #642 key=65570

n (2112)
Particle 150
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.339, 0.125, -0.191, 1.025)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.002366
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.002366
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 1 dE=1.146503
subgraph tracker simHits: 1 dE=1.146503
checkpointId: 0
x4@checkpoint: (114.821, 44.933, 2.765, 0.000)
p4@checkpoint: (0.356, 0.046, -0.195, 1.024)
raw GEN: n/a
raw SIM: SimTrack #643 key=65571

n (2112)
Particle 151
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.164, -0.050, -0.110, 0.961)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003600
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003600
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.418, -30.696, -12.470, 0.000)
p4@checkpoint: (-0.164, -0.050, -0.110, 0.961)
raw GEN: n/a
raw SIM: SimTrack #645 key=65577

gamma (22)
Particle 152
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.003, 0.004, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002419
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002419
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.477, -30.465, 228.389, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #646 key=65581

n (2112)
Particle 153
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.019, 0.060, 0.078, 0.945)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000162
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000162
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (22.452, 121.239, 103.278, 0.000)
p4@checkpoint: (-0.019, 0.037, -0.003, 0.941)
raw GEN: n/a
raw SIM: SimTrack #647 key=65582

gamma (22)
Particle 154
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.002, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002470
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002470
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-16.659, 122.169, -73.888, 0.000)
p4@checkpoint: (-0.000, 0.002, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #648 key=65583

gamma (22)
Particle 155
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000043
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000043
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.723, 115.662, -50.434, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #649 key=65586

gamma (22)
Particle 156
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.001, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000355
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000355
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (9.113, 74.967, 293.500, 0.000)
p4@checkpoint: (0.003, 0.001, 0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #650 key=65837

gamma (22)
Particle 157
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000269
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000269
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.229, 27.346, 197.276, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #651 key=65839

n (2112)
Particle 158
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.064, 0.008, -0.039, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 17 simE=0.003954
direct recHits: 0 missing=17 recoE=0.000000
subgraph simHits: 17 simE=0.003954
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.552, -123.069, 98.750, 0.000)
p4@checkpoint: (-0.025, -0.054, 0.028, 0.942)
raw GEN: n/a
raw SIM: SimTrack #653 key=65866

gamma (22)
Particle 159
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.003, 0.001, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003788
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003788
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (40.463, -116.472, 77.523, 0.000)
p4@checkpoint: (0.001, -0.003, 0.001, 0.004)
raw GEN: n/a
raw SIM: SimTrack #654 key=65918

gamma (22)
Particle 160
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (92.528, 81.496, 171.229, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #655 key=65923

pi0 (111)
Particle 161
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.317, 0.125, 0.070, 0.373)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 20 simE=0.375112
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #656 key=66302

Decay
Vertex 49
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.487, -51.504, 88.014, 7.249)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #367 key=71

n (2112)
Particle 162
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.011, 0.135, -0.104, 0.955)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000003
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000003
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.764, -69.620, -57.716, 0.000)
p4@checkpoint: (0.038, -0.017, -0.021, 0.941)
raw GEN: n/a
raw SIM: SimTrack #657 key=66303

gamma (22)
Particle 163
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003677
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003677
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (39.238, -116.890, -80.100, 0.000)
p4@checkpoint: (0.000, -0.001, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #658 key=66304

gamma (22)
Particle 164
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.240, 0.153, 0.053, 0.289)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.279934
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.279934
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.602, 13.098, 110.304, 0.000)
p4@checkpoint: (0.240, 0.153, 0.053, 0.289)
raw GEN: n/a
raw SIM: SimTrack #660 key=66329

gamma (22)
Particle 165
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.077, -0.028, 0.017, 0.084)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.095178
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.095178
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.325, -78.205, 104.171, 0.000)
p4@checkpoint: (0.077, -0.028, 0.017, 0.084)
raw GEN: n/a
raw SIM: SimTrack #661 key=66330

gamma (22)
Particle 166
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.132, -0.076, 0.002, 0.152)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.174448
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.174448
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-109.177, -57.299, 69.617, 0.000)
p4@checkpoint: (-0.132, -0.076, 0.002, 0.152)
raw GEN: n/a
raw SIM: SimTrack #662 key=66445

gamma (22)
Particle 167
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.543, -0.132, 0.216, 0.600)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.622751
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.622751
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.045, -23.473, 115.627, 0.000)
p4@checkpoint: (-0.543, -0.132, 0.216, 0.600)
raw GEN: n/a
raw SIM: SimTrack #663 key=66446

pi0 (111)
Particle 168
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.034, 0.065, -0.134, 0.203)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.051376
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #664 key=69540

Decay
Vertex 52
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.484, -6.211, -8.672, 1.203)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #370 key=74

n (2112)
Particle 169
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.173, -0.011, 0.106, 0.961)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.005801
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.005801
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 1 dE=0.907351
subgraph tracker simHits: 1 dE=0.907351
checkpointId: 0
x4@checkpoint: (-123.300, 0.091, 55.671, 0.000)
p4@checkpoint: (-0.169, 0.027, 0.110, 0.961)
raw GEN: n/a
raw SIM: SimTrack #665 key=69541

gamma (22)
Particle 170
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.004, -0.000, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006285
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006285
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-30.368, -119.502, -16.666, 0.000)
p4@checkpoint: (-0.000, -0.004, -0.000, 0.004)
raw GEN: n/a
raw SIM: SimTrack #666 key=69545

gamma (22)
Particle 171
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.009, 0.041, 0.020, 0.046)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.050873
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.050873
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-46.393, 114.239, 48.962, 0.000)
p4@checkpoint: (-0.009, 0.041, 0.020, 0.046)
raw GEN: n/a
raw SIM: SimTrack #667 key=69556

gamma (22)
Particle 172
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.025, 0.024, -0.153, 0.157)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000503
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000503
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-68.326, 37.668, -293.500, 0.000)
p4@checkpoint: (-0.025, 0.024, -0.153, 0.157)
raw GEN: n/a
raw SIM: SimTrack #668 key=69557

pi- (-211)
Particle 173
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.032, 0.084, -0.147, 0.222)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.003622
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 12 dE=3.862777
subgraph tracker simHits: 8 dE=3.862777
raw GEN: n/a
raw SIM: SimTrack #669 key=69833

HadronAtRest

Vertex 80
domain: SIM
reason: HadronAtRest
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.051, 14.701, -231.768, 17.385)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #402 key=106

pi- (-211)
Particle 174
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.254, -0.018, 0.260, 0.390)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 27 simE=0.019420
subgraph recHits: 0 missing=27 recoE=0.000000
direct tracker simHits: 35 dE=35.595367
subgraph tracker simHits: 10 dE=35.595367
raw GEN: n/a
raw SIM: SimTrack #670 key=69835

HadronInelastic

Vertex 78
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.687, -21.272, 186.358, 11.524)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #400 key=104

Bremsstrahlung
Vertex 79
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-112.100, 29.939, 177.059, 15.004)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #401 key=105

pi+ (211)
Particle 175
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.230, -0.187, 0.059, 0.333)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 6 dE=3.364253
subgraph tracker simHits: 4 dE=3.364253
raw GEN: n/a
raw SIM: SimTrack #671 key=69836

Decay
Vertex 77
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.549, 11.241, 116.347, 22.858)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #399 key=103

pdg(221)
Particle 176
pid: pdg(221)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.120, 0.157, -0.203, 0.617)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 39 simE=0.233325
subgraph recHits: 0 missing=39 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=7.922812
raw GEN: n/a
raw SIM: SimTrack #672 key=69837

Decay
Vertex 56
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.230, 14.051, 5.039, 2.242)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #375 key=79

n (2112)
Particle 177
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.258, 0.093, -0.278, 1.018)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 18 simE=0.010166
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #673 key=69838

HadronInelastic

Vertex 54
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-96.716, 45.974, -90.397, 13.886)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #372 key=76

HadronInelastic

Vertex 55
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-106.919, 46.798, -98.194, 15.941)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #374 key=78

n (2112)
Particle 178
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.036, 0.020, -0.011, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 15 simE=0.006597
direct recHits: 0 missing=15 recoE=0.000000
subgraph simHits: 15 simE=0.006597
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-110.915, 53.859, -94.697, 0.000)
p4@checkpoint: (-0.036, 0.020, -0.011, 0.941)
raw GEN: n/a
raw SIM: SimTrack #674 key=69844

gamma (22)
Particle 179
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.000, -0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003556
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003556
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (101.422, 70.117, -232.152, 0.000)
p4@checkpoint: (0.004, 0.000, -0.003, 0.005)
raw GEN: n/a
raw SIM: SimTrack #676 key=69861

n (2112)
Particle 180
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.037, 0.010, -0.003, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000013
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000013
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-114.559, 45.597, -98.451, 0.000)
p4@checkpoint: (-0.032, -0.009, -0.001, 0.940)
raw GEN: n/a
raw SIM: SimTrack #677 key=69862

pi0 (111)
Particle 181
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.025, 0.160, -0.003, 0.210)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 11 simE=0.166772
subgraph recHits: 0 missing=11 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #679 key=69870

Decay
Vertex 75
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.230, 14.051, 5.039, 2.242)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #397 key=101

pi0 (111)
Particle 182
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.149, -0.052, -0.097, 0.229)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 18 simE=0.022649
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=7.922812
raw GEN: n/a
raw SIM: SimTrack #680 key=69871

Decay
Vertex 57
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.230, 14.051, 5.039, 2.242)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #376 key=80

gamma (22)
Particle 183
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, 0.016, -0.131, 0.133)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.001811
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.001811
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-31.246, 49.579, -293.500, 0.000)
p4@checkpoint: (-0.010, 0.016, -0.131, 0.133)
raw GEN: n/a
raw SIM: SimTrack #681 key=69873

gamma (22)
Particle 184
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.034, 0.028, 0.045)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.042093
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.042093
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (11.781, 122.736, 94.137, 0.000)
p4@checkpoint: (0.006, 0.034, 0.028, 0.045)
raw GEN: n/a
raw SIM: SimTrack #682 key=69874

gamma (22)
Particle 185
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.118, -0.064, -0.008, 0.134)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 16 simE=0.021875
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 8 dE=6.494079
subgraph tracker simHits: 2 dE=7.922812
raw GEN: n/a
raw SIM: SimTrack #683 key=70049

PairConversion

Vertex 58
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (105.819, -47.675, -2.705, 6.575)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #377 key=81

Bremsstrahlung
Vertex 72
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (107.078, -44.637, -3.259, 6.787)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #394 key=98

Bremsstrahlung
Vertex 73
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.749, -44.973, -3.000, 6.811)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #395 key=99

Bremsstrahlung
Vertex 74
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.797, -45.017, -2.954, 6.813)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #396 key=100

gamma (22)
Particle 186
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.031, 0.011, -0.089, 0.095)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000774
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000774
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.362, 52.607, -293.500, 0.000)
p4@checkpoint: (0.031, 0.011, -0.089, 0.095)
raw GEN: n/a
raw SIM: SimTrack #684 key=70050

e+ (-11)
Particle 187
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.096, -0.052, -0.007, 0.110)
nProdVtx: 1 nDecayVtx: 13
nParents: 1 nChildren: 13
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.019089
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 2 dE=1.428733
subgraph tracker simHits: 2 dE=1.428733
checkpointId: 0
x4@checkpoint: (110.491, -54.724, -2.923, 0.000)
p4@checkpoint: (0.006, -0.097, -0.002, 0.098)
raw GEN: n/a
raw SIM: SimTrack #685 key=70164

Bremsstrahlung
Vertex 59
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (105.848, -47.691, -2.706, 6.576)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #378 key=82

Bremsstrahlung
Vertex 60
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.275, -47.946, -2.723, 6.593)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #379 key=83

Bremsstrahlung
Vertex 61
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.305, -47.965, -2.724, 6.594)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #380 key=84

Bremsstrahlung
Vertex 62
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.592, -48.139, -2.730, 6.605)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #382 key=86

Bremsstrahlung
Vertex 63
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.998, -50.397, 1.999, 8.308)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #385 key=89

Bremsstrahlung
Vertex 64
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.393, -51.599, 2.251, 8.370)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #386 key=90

Bremsstrahlung
Vertex 65
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (107.020, -52.498, 2.425, 8.407)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #387 key=91

Annihilation

Vertex 66
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.711, -58.250, 5.555, 8.906)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #388 key=92

Bremsstrahlung
Vertex 67
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.023, -54.326, 5.335, 9.132)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #389 key=93

Bremsstrahlung
Vertex 68
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (103.029, -54.332, 5.335, 9.132)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #390 key=94

Bremsstrahlung
Vertex 69
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.820, -54.581, 3.660, 9.536)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #391 key=95

Annihilation

Vertex 70
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (101.061, -55.841, 3.152, 9.642)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #392 key=96

Bremsstrahlung
Vertex 71
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (99.292, -59.092, 4.339, 8.988)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #393 key=97

gamma (22)
Particle 188
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000838
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000838
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.084, -51.382, -3.062, 0.000)
p4@checkpoint: (0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #686 key=70165

gamma (22)
Particle 189
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, -0.003, -0.000, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004430
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004430
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.024, -51.511, -2.863, 0.000)
p4@checkpoint: (0.005, -0.003, -0.000, 0.005)
raw GEN: n/a
raw SIM: SimTrack #687 key=70166

gamma (22)
Particle 190
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001195
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001195
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.037, -51.483, -2.849, 0.000)
p4@checkpoint: (0.001, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #688 key=70167

gamma (22)
Particle 191
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000087
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000087
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.018, -51.524, -2.845, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #690 key=70169

gamma (22)
Particle 192
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000608
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000608
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.804, -54.086, 2.599, 0.000)
p4@checkpoint: (0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #693 key=70181

gamma (22)
Particle 193
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000208
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000208
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.941, -55.819, 3.192, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #694 key=70182

gamma (22)
Particle 194
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, -0.007, 0.001, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.007613
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.007613
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.487, -56.705, 3.142, 0.000)
p4@checkpoint: (0.004, -0.007, 0.001, 0.008)
raw GEN: n/a
raw SIM: SimTrack #695 key=70183

gamma (22)
Particle 195
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001405
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001405
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.915, -70.845, 5.913, 0.000)
p4@checkpoint: (0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #696 key=70204

gamma (22)
Particle 196
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000911
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000911
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.957, -59.567, 5.292, 0.000)
p4@checkpoint: (0.001, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #697 key=70211

gamma (22)
Particle 197
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000062
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000062
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.919, -59.636, 5.008, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #698 key=70212

gamma (22)
Particle 198
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000437
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000437
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.463, -60.453, 4.495, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #699 key=70215

gamma (22)
Particle 199
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000206
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000206
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.767, -65.014, 6.384, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #700 key=70224

gamma (22)
Particle 200
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001089
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001089
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.910, -59.652, 6.800, 0.000)
p4@checkpoint: (0.001, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #701 key=70231

gamma (22)
Particle 201
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.000, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001007
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001007
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-90.650, -83.579, -67.999, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #702 key=70248

gamma (22)
Particle 202
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001063
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001063
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.311, -50.883, 9.431, 0.000)
p4@checkpoint: (0.000, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #703 key=70257

gamma (22)
Particle 203
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000716
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000716
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.569, -54.565, 7.070, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #704 key=70258

gamma (22)
Particle 204
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.028, -0.017, -0.009, 0.034)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 2 simE=0.006956
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 4 simE=0.007307
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-112.756, -49.890, -28.090, 0.000)
p4@checkpoint: (-0.028, -0.017, -0.009, 0.034)
raw GEN: n/a
raw SIM: SimTrack #705 key=70264

Annihilation

Vertex 76
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.155, 12.646, -89.865, 15.052)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #398 key=102

gamma (22)
Particle 205
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.177, 0.006, 0.177)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.159465
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.159465
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-6.447, 123.131, 8.852, 0.000)
p4@checkpoint: (0.003, 0.177, 0.006, 0.177)
raw GEN: n/a
raw SIM: SimTrack #706 key=70265

gamma (22)
Particle 206
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000211
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000211
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.665, 20.016, -99.128, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #707 key=70318

gamma (22)
Particle 207
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.002, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000140
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000140
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (118.552, 33.888, -84.574, 0.000)
p4@checkpoint: (0.000, 0.002, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #708 key=70319

mu+ (-13)
Particle 208
caloBoundary
pid: mu+ (-13)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.177, -0.057, 0.066, 0.224)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-18.707, 1.358, 293.500, 0.000)
p4@checkpoint: (0.088, -0.069, 0.035, 0.158)
raw GEN: n/a
raw SIM: SimTrack #709 key=70397

n (2112)
Particle 209
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.051, -0.265, -0.246, 1.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.012461
direct recHits: 0 missing=23 recoE=0.000000
subgraph simHits: 23 simE=0.012461
subgraph recHits: 0 missing=23 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-43.388, -115.414, 95.943, 0.000)
p4@checkpoint: (-0.075, -0.251, -0.255, 1.008)
raw GEN: n/a
raw SIM: SimTrack #711 key=70544

n (2112)
Particle 210
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.118, 0.024, -0.010, 0.947)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000133
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000133
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.012, 8.428, 173.747, 0.000)
p4@checkpoint: (0.118, 0.024, -0.010, 0.947)
raw GEN: n/a
raw SIM: SimTrack #712 key=70548

gamma (22)
Particle 211
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.006, 0.004, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.006513
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.006513
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-17.194, -122.095, 249.562, 0.000)
p4@checkpoint: (0.000, -0.006, 0.004, 0.007)
raw GEN: n/a
raw SIM: SimTrack #713 key=70549

gamma (22)
Particle 212
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000215
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000215
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-73.398, -99.074, 253.572, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #716 key=70552

gamma (22)
Particle 213
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000098
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000098
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-118.788, 33.051, 174.787, 0.000)
p4@checkpoint: (-0.002, 0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #717 key=70575

gamma (22)
Particle 214
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.000, 0.005, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003622
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003622
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.464, -14.332, 83.410, 0.000)
p4@checkpoint: (-0.002, -0.000, 0.005, 0.005)
raw GEN: n/a
raw SIM: SimTrack #719 key=70876

gamma (22)
Particle 215
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.002, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001216
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001216
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.204, -105.869, 39.013, 0.000)
p4@checkpoint: (0.000, -0.002, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #720 key=71276

n (2112)
Particle 216
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.054, 0.029, -0.020, 0.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.003441
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.003441
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.233, 113.475, 96.967, 0.000)
p4@checkpoint: (0.054, 0.029, -0.020, 0.942)
raw GEN: n/a
raw SIM: SimTrack #721 key=71277

n (2112)
Particle 217
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.030, -0.075, 0.010, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000018
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000018
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-121.804, -19.147, 213.845, 0.000)
p4@checkpoint: (-0.030, -0.075, 0.010, 0.943)
raw GEN: n/a
raw SIM: SimTrack #722 key=71575

p (2212)
Particle 218
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.775, -0.505, 0.481, 1.402)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.001705
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.001705
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 10 dE=10.492152
subgraph tracker simHits: 10 dE=10.492152
checkpointId: 0
x4@checkpoint: (-0.551, -29.557, 293.500, 0.000)
p4@checkpoint: (-0.771, -0.355, 0.573, 1.389)
raw GEN: n/a
raw SIM: SimTrack #723 key=72178

gamma (22)
Particle 219
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.005, 0.002, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000215
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000215
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (3.090, -103.353, 293.500, 0.000)
p4@checkpoint: (-0.000, -0.005, 0.002, 0.006)
raw GEN: n/a
raw SIM: SimTrack #724 key=72181

pi+ (211)
Particle 220
caloBoundary
pid: pi+ (211)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.087, -0.014, 0.598, 1.249)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 76 simE=1.045971
direct recHits: 0 missing=76 recoE=0.000000
subgraph simHits: 77 simE=1.046181
subgraph recHits: 0 missing=77 recoE=0.000000
direct tracker simHits: 22 dE=22.995939
subgraph tracker simHits: 12 dE=22.995937
checkpointId: 0
x4@checkpoint: (-106.792, 61.631, 83.958, 0.000)
p4@checkpoint: (-0.452, 0.970, 0.579, 1.225)
raw GEN: n/a
raw SIM: SimTrack #726 key=73102

Annihilation

Vertex 103
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.348, 73.895, 53.040, 6.685)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #433 key=137

n (2112)
Particle 221
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.238, 0.538, 0.340, 1.159)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.046910
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.046910
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-63.438, 105.728, 86.740, 0.000)
p4@checkpoint: (-0.238, 0.538, 0.340, 1.159)
raw GEN: n/a
raw SIM: SimTrack #727 key=73103

pi0 (111)
Particle 222
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.115, 0.047, 0.227, 0.292)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 21 simE=0.094394
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 3 dE=4.102146
raw GEN: n/a
raw SIM: SimTrack #728 key=73104

Decay
Vertex 85
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.710, 6.949, 24.218, 1.184)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #407 key=111

gamma (22)
Particle 223
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001196
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001196
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-71.089, 100.744, 9.016, 0.000)
p4@checkpoint: (-0.001, 0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #730 key=73106

gamma (22)
Particle 224
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.008, 0.017, 0.131, 0.133)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.002268
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 2 dE=2.000000
subgraph tracker simHits: 3 dE=4.102146
raw GEN: n/a
raw SIM: SimTrack #731 key=73119

PairConversion

Vertex 98
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.005, 7.541, 28.791, 1.338)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #422 key=126

gamma (22)
Particle 225
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.124, 0.030, 0.096, 0.160)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 16 simE=0.092126
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #732 key=73120

PairConversion

Vertex 86
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.591, 36.714, 120.463, 6.521)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #408 key=112

Annihilation

Vertex 87
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (104.484, 35.249, 146.153, 9.084)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #409 key=113

e- (11)
Particle 226
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.104, 0.026, 0.081, 0.134)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 7
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 15 simE=0.091916
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #733 key=73121

Bremsstrahlung
Vertex 88
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.564, 37.436, 122.021, 6.609)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #410 key=114

Bremsstrahlung
Vertex 89
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (106.693, 37.499, 122.126, 6.615)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #411 key=115

Bremsstrahlung
Vertex 90
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.703, 39.372, 124.302, 6.732)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #412 key=116

Bremsstrahlung
Vertex 91
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.783, 39.486, 124.449, 6.739)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #413 key=117

Bremsstrahlung
Vertex 92
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.802, 39.511, 124.480, 6.740)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #414 key=118

Bremsstrahlung
Vertex 93
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.825, 39.546, 124.517, 6.742)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #415 key=119

Bremsstrahlung
Vertex 94
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (109.175, 40.165, 125.572, 6.785)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #416 key=120

gamma (22)
Particle 227
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000210
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000210
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.455, 48.279, 149.930, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #734 key=73129

gamma (22)
Particle 228
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.023, 0.012, 0.019, 0.032)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.034679
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.034679
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.902, 42.067, 129.705, 0.000)
p4@checkpoint: (0.023, 0.012, 0.019, 0.032)
raw GEN: n/a
raw SIM: SimTrack #735 key=73132

gamma (22)
Particle 229
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.003, 0.005, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008035
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008035
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.815, 42.307, 129.954, 0.000)
p4@checkpoint: (0.006, 0.003, 0.005, 0.008)
raw GEN: n/a
raw SIM: SimTrack #736 key=73134

gamma (22)
Particle 230
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.004, 0.007, 0.009, 0.012)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 1 simE=0.000081
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 4 simE=0.005078
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (105.191, 64.325, 110.611, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #737 key=73138

Bremsstrahlung
Vertex 95
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.715, 39.394, 124.327, 6.733)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #419 key=123

Bremsstrahlung
Vertex 96
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.722, 39.403, 124.340, 6.734)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #420 key=124

Bremsstrahlung
Vertex 97
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (108.815, 39.609, 124.461, 6.743)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #421 key=125

gamma (22)
Particle 231
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.010, 0.014, 0.019, 0.026)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.036406
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.036406
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (114.001, 46.974, 134.300, 0.000)
p4@checkpoint: (0.010, 0.014, 0.019, 0.026)
raw GEN: n/a
raw SIM: SimTrack #738 key=73140

gamma (22)
Particle 232
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000799
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000799
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.793, 47.477, 134.073, 0.000)
p4@checkpoint: (0.000, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #739 key=73141

gamma (22)
Particle 233
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003550
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003550
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.342, 48.544, 132.186, 0.000)
p4@checkpoint: (0.001, 0.003, 0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #740 key=73142

gamma (22)
Particle 234
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003369
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003369
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.469, 50.533, 143.562, 0.000)
p4@checkpoint: (0.000, 0.001, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #741 key=73144

gamma (22)
Particle 235
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000310
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000310
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (117.371, 37.776, 137.980, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #744 key=73186

gamma (22)
Particle 236
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.003, 0.004, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004559
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004559
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.934, 47.137, 134.140, 0.000)
p4@checkpoint: (0.002, 0.003, 0.004, 0.005)
raw GEN: n/a
raw SIM: SimTrack #745 key=73187

gamma (22)
Particle 237
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000128
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000128
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.971, 41.875, 119.509, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #746 key=73193

e- (11)
Particle 238
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.004, 0.009, 0.061, 0.062)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=1.036358
raw GEN: n/a
raw SIM: SimTrack #747 key=73211

Bremsstrahlung
Vertex 102
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-22.500, 6.104, 72.071, 2.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #430 key=134

e+ (-11)
Particle 239
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.005, 0.009, 0.070, 0.071)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.002268
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=1.065789
raw GEN: n/a
raw SIM: SimTrack #748 key=73212

Bremsstrahlung
Vertex 99
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.303, 8.903, 41.585, 1.769)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #424 key=128

Annihilation

Vertex 100
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.148, 8.345, 99.433, 3.877)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #427 key=131

Bremsstrahlung
Vertex 101
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-18.968, 9.090, 74.326, 2.882)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #429 key=133

gamma (22)
Particle 240
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001008
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001008
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-62.196, 106.464, 212.899, 0.000)
p4@checkpoint: (-0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #750 key=73214

gamma (22)
Particle 241
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000339
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000339
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=1.065789
subgraph tracker simHits: 1 dE=1.065789
checkpointId: 0
x4@checkpoint: (111.158, -53.356, -39.607, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #753 key=73230

gamma (22)
Particle 242
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-68.839, 102.294, 144.989, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #754 key=73231

gamma (22)
Particle 243
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000410
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000410
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.700, -25.186, 160.520, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #756 key=73237

gamma (22)
Particle 244
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.006, 0.021, 0.022)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=1.036358
subgraph tracker simHits: 1 dE=1.036358
checkpointId: 0
x4@checkpoint: (-1.662, -9.037, 293.500, 0.000)
p4@checkpoint: (-0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #757 key=73249

gamma (22)
Particle 245
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000210
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000210
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.460, -65.506, -82.505, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #760 key=73333

pi- (-211)
Particle 246
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.187, 0.077, 8.598, 8.601)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.511, 4.462, 482.992, 0.000)
p4@checkpoint: (-0.198, 0.038, 8.597, 8.601)
raw GEN: n/a
raw SIM: SimTrack #761 key=73577

p (2212)
Particle 247
caloBoundary
pid: p (2212)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.321, 0.400, 35.194, 35.210)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=6.000000
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=6.000000
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-10.610, 12.653, 1114.950, 0.000)
p4@checkpoint: (-0.315, 0.401, 35.192, 35.208)
raw GEN: n/a
raw SIM: SimTrack #762 key=73578

gamma (22)
Particle 248
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000346
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000346
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-117.042, -38.782, -100.752, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #765 key=76081

gamma (22)
Particle 249
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001234
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001234
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.214, 16.329, -39.999, 0.000)
p4@checkpoint: (0.001, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #766 key=76082

pi0 (111)
Particle 250
pid: pi0 (111)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.214, 0.014, -0.110, 0.276)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.250646
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #767 key=76083

Decay
Vertex 107
domain: SIM
reason: Decay
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.868, -13.458, -4.447, 2.477)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #440 key=144

n (2112)
Particle 251
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.145, -0.030, -0.042, 0.952)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.000684
direct recHits: 0 missing=10 recoE=0.000000
subgraph simHits: 10 simE=0.000684
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-118.445, -34.259, -33.816, 0.000)
p4@checkpoint: (-0.145, -0.030, -0.042, 0.952)
raw GEN: n/a
raw SIM: SimTrack #768 key=76085

gamma (22)
Particle 252
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, -0.000, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002753
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002753
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (75.643, -97.371, 18.365, 0.000)
p4@checkpoint: (0.002, -0.002, 0.000, 0.003)
raw GEN: n/a
raw SIM: SimTrack #769 key=76086

gamma (22)
Particle 253
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001547
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001547
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-116.544, -40.253, 83.590, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #772 key=76148

gamma (22)
Particle 254
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.036, -0.032, 0.019, 0.052)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 6 simE=0.042484
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 8 simE=0.048604
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.808, -80.019, 34.668, 0.000)
p4@checkpoint: (-0.024, -0.017, 0.011, 0.031)
raw GEN: n/a
raw SIM: SimTrack #773 key=76214

Bremsstrahlung
Vertex 108
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-88.538, -76.694, 32.518, 5.874)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #441 key=145

Bremsstrahlung
Vertex 109
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.612, -77.176, 34.478, 6.011)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #442 key=146

Bremsstrahlung
Vertex 110
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-86.987, -77.580, 35.090, 6.040)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #443 key=147

gamma (22)
Particle 255
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.178, 0.046, -0.129, 0.224)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.202042
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.202042
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.530, 13.756, -79.988, 0.000)
p4@checkpoint: (-0.178, 0.046, -0.129, 0.224)
raw GEN: n/a
raw SIM: SimTrack #774 key=76215

gamma (22)
Particle 256
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.003, 0.001, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001227
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001227
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-93.610, -80.250, 33.530, 0.000)
p4@checkpoint: (-0.001, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #775 key=76291

gamma (22)
Particle 257
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.002, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.004696
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.004696
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-92.908, -81.061, 42.073, 0.000)
p4@checkpoint: (-0.002, -0.002, 0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #776 key=76293

gamma (22)
Particle 258
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000197
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000197
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-95.452, -78.051, 50.156, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #777 key=76295

mu+ (-13)
Particle 259
caloBoundary
pid: mu+ (-13)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.073, 0.070, 0.168, 0.223)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.745794
subgraph tracker simHits: 2 dE=0.745794
checkpointId: 0
x4@checkpoint: (26.016, -12.956, 293.500, 0.000)
p4@checkpoint: (0.029, -0.000, 0.105, 0.152)
raw GEN: n/a
raw SIM: SimTrack #778 key=76326

pi- (-211)
Particle 260
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.402, 0.755, -10.794, 10.829)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=2.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=2.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-6.558, 5.341, -420.645, 0.000)
p4@checkpoint: (-0.480, 0.723, -10.783, 10.819)
raw GEN: n/a
raw SIM: SimTrack #779 key=80550

pi- (-211)
Particle 261
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.111, -0.621, -10.141, 10.161)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-0.587, -8.401, -412.557, 0.000)
p4@checkpoint: (0.182, -0.609, -10.109, 10.130)
raw GEN: n/a
raw SIM: SimTrack #781 key=80553

n (2112)
Particle 262
caloBoundary
pid: n (2112)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.250, 0.042, -0.059, 0.975)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.007682
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.007682
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (7.876, -0.530, -313.681, 0.000)
p4@checkpoint: (0.250, 0.042, -0.059, 0.975)
raw GEN: n/a
raw SIM: SimTrack #786 key=80559

pi- (-211)
Particle 263
caloBoundary
pid: pi- (-211)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.045, -0.068, -1.804, 1.811)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 3 dE=3.000000
subgraph tracker simHits: 3 dE=3.000000
checkpointId: 0
x4@checkpoint: (9.027, -3.304, -293.500, 0.000)
p4@checkpoint: (0.080, 0.025, -1.795, 1.802)
raw GEN: n/a
raw SIM: SimTrack #795 key=82079

mu- (13)
Particle 264
caloBoundary
pid: mu- (13)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.038, -0.157, -2.487, 2.494)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 31 dE=31.000000
subgraph tracker simHits: 31 dE=31.000000
checkpointId: 0
x4@checkpoint: (7.032, -15.773, -293.500, 0.000)
p4@checkpoint: (0.132, -0.078, -2.462, 2.469)
raw GEN: n/a
raw SIM: SimTrack #797 key=82190

e- (11)
Particle 265
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.411, 0.167, 1.876, 2.354)
nProdVtx: 1 nDecayVtx: 13
nParents: 1 nChildren: 13
nCheckpoints: 1
direct simHits: 40 simE=1.518292
direct recHits: 0 missing=40 recoE=0.000000
subgraph simHits: 95 simE=1.916723
subgraph recHits: 0 missing=95 recoE=0.000000
direct tracker simHits: 2 dE=2.181993
subgraph tracker simHits: 2 dE=2.181993
checkpointId: 0
x4@checkpoint: (121.646, 20.126, 167.178, 0.000)
p4@checkpoint: (0.856, 0.420, 1.279, 1.595)
raw GEN: n/a
raw SIM: SimTrack #799 key=130596

Bremsstrahlung
Vertex 128
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (88.023, 10.447, 120.523, 5.096)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #472 key=176

Bremsstrahlung
Vertex 129
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.266, 16.918, 156.389, 6.595)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #473 key=177

Bremsstrahlung
Vertex 130
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.307, 16.933, 156.447, 6.598)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #474 key=178

Bremsstrahlung
Vertex 131
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.364, 16.955, 156.527, 6.601)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #475 key=179

Bremsstrahlung
Vertex 132
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.402, 16.970, 156.583, 6.603)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #476 key=180

Bremsstrahlung
Vertex 133
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.025, 17.211, 157.472, 6.640)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #477 key=181

Bremsstrahlung
Vertex 134
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (117.054, 18.035, 160.410, 6.763)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #478 key=182

Bremsstrahlung
Vertex 135
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.434, 16.962, 156.609, 6.605)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #479 key=183

Annihilation

Vertex 145
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.254, 17.114, 156.652, 6.607)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #489 key=193

Bremsstrahlung
Vertex 146
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.273, 16.891, 156.390, 6.595)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #490 key=194

Bremsstrahlung
Vertex 147
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.334, 16.990, 156.566, 6.602)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #491 key=195

Bremsstrahlung
Vertex 148
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.376, 16.975, 156.841, 6.611)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #492 key=196

Bremsstrahlung
Vertex 149
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.419, 16.738, 157.013, 6.622)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #493 key=197

e+ (-11)
Particle 266
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.184, 0.022, 0.245, 0.307)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 4 simE=0.000591
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 18 simE=0.012063
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 21 dE=20.871843
subgraph tracker simHits: 13 dE=28.995354
checkpointId: 0
x4@checkpoint: (87.564, 7.032, 293.500, 0.000)
p4@checkpoint: (0.098, 0.066, 0.148, 0.189)
raw GEN: n/a
raw SIM: SimTrack #800 key=130597

Bremsstrahlung
Vertex 116
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (88.010, 10.445, 120.506, 5.095)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #456 key=160

Bremsstrahlung
Vertex 117
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (95.348, 9.644, 130.365, 5.507)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #457 key=161

Bremsstrahlung
Vertex 118
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.981, 8.379, 134.226, 5.669)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #458 key=162

gamma (22)
Particle 267
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000933
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000933
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.379, 15.042, 165.977, 0.000)
p4@checkpoint: (0.001, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #801 key=130598

gamma (22)
Particle 268
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.001, 0.005, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.005701
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.005701
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.299, -0.565, 169.594, 0.000)
p4@checkpoint: (0.003, -0.001, 0.005, 0.006)
raw GEN: n/a
raw SIM: SimTrack #802 key=130600

gamma (22)
Particle 269
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.041, -0.025, 0.063, 0.079)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 9 simE=0.004838
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 3 dE=8.123512
raw GEN: n/a
raw SIM: SimTrack #803 key=130602

PairConversion

Vertex 119
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.265, -1.012, 157.753, 6.656)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #461 key=165

Bremsstrahlung
Vertex 125
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.364, -1.089, 157.914, 6.663)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #469 key=173

Bremsstrahlung
Vertex 126
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.386, -1.103, 157.942, 6.664)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #470 key=174

Bremsstrahlung
Vertex 127
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.388, 0.814, 163.640, 6.911)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #471 key=175

e+ (-11)
Particle 270
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.027, -0.016, 0.040, 0.051)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 5 simE=0.001018
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=5.830309
raw GEN: n/a
raw SIM: SimTrack #806 key=130880

Bremsstrahlung
Vertex 120
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.253, -1.655, 176.936, 7.515)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #464 key=168

Bremsstrahlung
Vertex 121
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.177, -4.083, 183.080, 7.744)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #465 key=169

Annihilation

Vertex 122
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.321, -3.764, 188.570, 8.063)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #466 key=170

gamma (22)
Particle 271
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000184
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000184
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.911, -9.792, 190.406, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #809 key=130885

gamma (22)
Particle 272
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.007, -0.002, 0.009, 0.011)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000610
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 6 dE=5.830309
subgraph tracker simHits: 2 dE=5.830309
checkpointId: 0
x4@checkpoint: (84.040, -0.634, 293.500, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #810 key=130889

Compton
Vertex 124
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (84.096, -11.464, 224.645, 9.454)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #468 key=172

gamma (22)
Particle 273
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1 simE=0.000163
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 2 simE=0.000224
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (113.582, 47.980, 209.109, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #811 key=130892

Bremsstrahlung
Vertex 123
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.393, 70.383, 214.671, 10.769)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #467 key=171

gamma (22)
Particle 274
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000061
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000061
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.964, 72.182, 213.493, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #812 key=130894

e- (11)
Particle 275
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.007, -0.002, 0.007, 0.010)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000610
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000610
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (84.753, -12.290, 293.500, 0.000)
p4@checkpoint: (0.001, -0.006, 0.006, 0.009)
raw GEN: n/a
raw SIM: SimTrack #813 key=130896

gamma (22)
Particle 276
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002268
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002268
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.138, -6.319, 170.187, 0.000)
p4@checkpoint: (0.001, -0.001, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #814 key=130915

gamma (22)
Particle 277
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.001380
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.001380
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.101, -6.998, 170.356, 0.000)
p4@checkpoint: (0.001, -0.001, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #815 key=130916

gamma (22)
Particle 278
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000172
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000172
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=2.293204
subgraph tracker simHits: 2 dE=2.293204
checkpointId: 0
x4@checkpoint: (110.933, 21.184, 293.500, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #816 key=130917

gamma (22)
Particle 279
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.146, 0.018, 0.194, 0.243)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.045163
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #817 key=130933

PairConversion

Vertex 150
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.996, 13.698, 156.361, 6.596)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #494 key=198

Bremsstrahlung
Vertex 155
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.102, 13.714, 156.505, 6.602)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #499 key=203

Bremsstrahlung
Vertex 156
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.120, 16.821, 159.204, 7.045)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #500 key=204

gamma (22)
Particle 280
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.003, 0.010, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.011979
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.011979
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.713, 19.719, 166.924, 0.000)
p4@checkpoint: (0.007, 0.003, 0.010, 0.013)
raw GEN: n/a
raw SIM: SimTrack #818 key=130939

gamma (22)
Particle 281
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001279
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001279
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.096, 17.190, 159.644, 0.000)
p4@checkpoint: (0.001, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #819 key=130940

gamma (22)
Particle 282
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.024, 0.009, 0.034, 0.042)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.011486
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.011486
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.705, 19.767, 166.935, 0.000)
p4@checkpoint: (0.024, 0.009, 0.034, 0.042)
raw GEN: n/a
raw SIM: SimTrack #820 key=130941

gamma (22)
Particle 283
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.225, 0.086, 0.319, 0.400)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 37 simE=0.321640
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #821 key=130942

PairConversion

Vertex 136
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.500, 17.007, 156.721, 6.609)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #480 key=184

Bremsstrahlung
Vertex 137
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.260, 15.908, 169.669, 7.076)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #481 key=185

Bremsstrahlung
Vertex 138
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.274, 15.910, 169.734, 7.079)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #482 key=186

Bremsstrahlung
Vertex 139
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.691, 15.147, 174.273, 7.335)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #483 key=187

Bremsstrahlung
Vertex 140
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.585, 15.219, 174.065, 7.347)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #484 key=188

gamma (22)
Particle 284
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000105
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000105
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.518, 26.043, 162.555, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #822 key=130945

gamma (22)
Particle 285
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001054
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001054
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.670, 19.980, 167.149, 0.000)
p4@checkpoint: (0.001, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #823 key=130947

gamma (22)
Particle 286
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001505
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001505
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.542, 13.651, 149.288, 0.000)
p4@checkpoint: (0.001, -0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #824 key=131933

e- (11)
Particle 287
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.199, 0.079, 0.288, 0.359)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 32 simE=0.318157
direct recHits: 0 missing=32 recoE=0.000000
subgraph simHits: 33 simE=0.320639
subgraph recHits: 0 missing=33 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.345, 21.871, 168.514, 0.000)
p4@checkpoint: (0.132, 0.152, 0.284, 0.348)
raw GEN: n/a
raw SIM: SimTrack #825 key=131934

Bremsstrahlung
Vertex 141
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (114.535, 17.021, 156.772, 6.611)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #485 key=189

Bremsstrahlung
Vertex 142
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.052, 17.245, 157.551, 6.643)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #486 key=190

Bremsstrahlung
Vertex 143
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (117.744, 18.720, 161.756, 6.817)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #487 key=191

Bremsstrahlung
Vertex 144
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (117.851, 18.791, 161.936, 6.824)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #488 key=192

gamma (22)
Particle 288
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000240
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000240
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.452, 21.268, 222.589, 0.000)
p4@checkpoint: (0.000, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #826 key=131942

gamma (22)
Particle 289
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000116
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000116
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.003, 17.836, 180.015, 0.000)
p4@checkpoint: (0.001, 0.000, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #827 key=131943

gamma (22)
Particle 290
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000493
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000493
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-107.192, -60.932, 106.316, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #828 key=131953

gamma (22)
Particle 291
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000153
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000153
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-1.947, -123.285, 147.658, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #829 key=131954

gamma (22)
Particle 292
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001644
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001644
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.669, 19.988, 167.395, 0.000)
p4@checkpoint: (0.001, 0.000, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #830 key=131961

gamma (22)
Particle 293
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000540
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000540
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.640, 20.163, 167.357, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #831 key=131962

gamma (22)
Particle 294
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000130
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000130
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.471, 21.160, 167.924, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #832 key=131963

gamma (22)
Particle 295
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000167
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000167
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.460, 21.222, 168.083, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #833 key=131964

gamma (22)
Particle 296
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000196
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000196
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.963, 9.116, 155.181, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #835 key=132225

gamma (22)
Particle 297
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.001, 0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003264
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003264
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.767, 19.380, 168.469, 0.000)
p4@checkpoint: (0.002, 0.001, 0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #836 key=132228

gamma (22)
Particle 298
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000254
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000254
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.484, 21.083, 159.965, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #837 key=132229

gamma (22)
Particle 299
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000326
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000326
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.816, -29.102, 221.065, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #838 key=132233

gamma (22)
Particle 300
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000182
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000182
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.040, 8.006, 151.601, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #839 key=132239

e+ (-11)
Particle 301
pid: e+ (-11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.120, 0.014, 0.160, 0.201)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 3 simE=0.002683
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 10 simE=0.038005
subgraph recHits: 0 missing=10 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.670, 12.449, 166.994, 0.000)
p4@checkpoint: (0.075, -0.052, 0.122, 0.152)
raw GEN: n/a
raw SIM: SimTrack #840 key=132249

Bremsstrahlung
Vertex 151
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (115.601, 13.782, 157.175, 6.630)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #495 key=199

Bremsstrahlung
Vertex 152
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (116.963, 13.940, 158.971, 6.705)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #496 key=200

Bremsstrahlung
Vertex 153
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (117.046, 9.558, 207.877, 8.746)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #497 key=201

Bremsstrahlung
Vertex 154
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (117.973, 10.120, 209.273, 8.805)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #498 key=202

gamma (22)
Particle 302
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.021, 0.003, 0.028, 0.035)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.033700
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.033700
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.420, 14.701, 166.448, 0.000)
p4@checkpoint: (0.021, 0.003, 0.028, 0.035)
raw GEN: n/a
raw SIM: SimTrack #841 key=132251

gamma (22)
Particle 303
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000650
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000650
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.448, 14.467, 166.287, 0.000)
p4@checkpoint: (0.000, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #842 key=132252

gamma (22)
Particle 304
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000532
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000532
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.531, 13.754, 216.914, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #843 key=132266

gamma (22)
Particle 305
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000439
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000439
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.704, 12.107, 216.026, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #844 key=132271

gamma (22)
Particle 306
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.000, 0.006, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.006560
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.006560
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.462, 14.353, 168.354, 0.000)
p4@checkpoint: (0.003, 0.000, 0.006, 0.007)
raw GEN: n/a
raw SIM: SimTrack #845 key=132411

gamma (22)
Particle 307
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000599
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000599
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.282, 2.097, 165.922, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #846 key=132420

connector

Particle 308
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 701 simE=69.180542
subgraph recHits: 0 missing=701 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 54 dE=102.202339
raw GEN: n/a
raw SIM: n/a

initial state

Vertex 158
initial state (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: n/a

connector

Particle 309
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 83
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1766 simE=55.110054
subgraph recHits: 0 missing=1766 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 594 dE=1571.661987
raw GEN: n/a
raw SIM: n/a

underlying event
Vertex 159
underlying event (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 1 nOut: 83
raw GEN: n/a
raw SIM: n/a

interaction

Vertex 157
interaction (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: yes isSink: no
x4: (0.000, 0.000, 3.451, 0.198)
nIn: 0 nOut: 2
raw GEN: n/a
raw SIM: n/a